HK IMO Training 2018-2019
Problem Set 2
Date due: 29 September 2018

Problem 9. Let P be the intersection point of the diagonals AC and BD in a cyclic quadrilateral $ABCD$. A circle through P touches the side CD in the midpoint M of this side and intersects the segments BD and AC in the points Q and R, respectively. Let S be a point on the segment BD such that $BS = DQ$. The parallel to AB through S intersects AC at T. Prove that $AT = RC$.

Problem 10. Find all positive integers n such that the decimal representation of n^2 consists of odd digits only.

Problem 11. The real numbers $x_1, x_2, \ldots, x_{2019}$ satisfy

$$x_1 + x_2 = 2y_1, \quad x_2 + x_3 = 2y_2, \quad \ldots, \quad x_{2019} + x_1 = 2y_{2019}$$

where $y_1, y_2, \ldots, y_{2019}$ is a permutation of $x_1, x_2, \ldots, x_{2019}$. Prove that $x_1 = x_2 = \ldots = x_{2019}$.

Problem 12. There are $2n$ different numbers in a row. By one move we can interchange any two numbers or interchange any three numbers cyclically (choose a, b, c and place a instead of b, b instead of c and c instead of a). What is the minimum number of moves that is always sufficient to arrange the numbers in increasing order?